Run report: ctrlK-q3-cone-run-20261024T2151Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (exhaustive, q = 3) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: ctrlK-q3-cone 20261024T2151Z
git HEAD: 0fb870d0ee76ba6a1b059e9e34c869518fb412b0
tool: examples/genus4_n4_census.rs sha256 f4c7f7355c838b6f112e09cb412fd3353d087309962c64cd39ad49d79f4c9dad
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 a95b1e35748cefe23a044d8cfd9e059b723cf71fdbc4af56db71257da42c8b33
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/ctrlK-q3-cone-run-20261024T2151Z CENSUS 3 cone exh 1 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 48
```

tool_log.txt

```
cone q=3: samples=14348907 (exhaustive) res_shaped=177363 allowed_trace_candidates=22464 smooth_prime_order_hits=8208 smooth_fraction_among_prime_cands=0.3654
smooth_fraction_other_res_shaped(sub-sample 2000)=955/2000 secs=48.1
cone q=3: total prime-order mass estimate 6.333 +- 0.070 (hit rate per sampled form 5.720e-4)
```

summary.txt

```
cone q=3: samples=14348907 (exhaustive) res_shaped=177363 allowed_trace_candidates=22464 smooth_prime_order_hits=8208 smooth_fraction_among_prime_cands=0.3654
smooth_fraction_other_res_shaped(sub-sample 2000)=955/2000 secs=48.1
cone q=3: total prime-order mass estimate 6.333 +- 0.070 (hit rate per sampled form 5.720e-4)
```

cand_t_hist.csv (first 12 of 26 rows)

type	t	candidates
cone	−18	2160
cone	−17	648
cone	−16	1296
cone	−14	6480
cone	−13	2592
cone	−12	1296
cone	−11	2592
cone	−10	1296
cone	−9	5616
cone	−8	16200
cone	−7	3888
cone	−6	2592

hits.csv (first 12 of 22464 rows)

type	code	t	nE	smooth
cone	22825	−1	83	false
cone	23152	−1	83	false
cone	23483	−1	83	false
cone	23891	−1	83	false

cone	24310	−7	89	true
cone	24320	−7	89	true
cone	24682	−1	83	false
cone	24764	−1	83	false
cone	25286	−1	83	false
cone	26023	−1	83	false
cone	28615	−1	83	false
cone	29132	−1	83	false

mass_by_t.csv (first 3 of 3 rows)

type	t	nE	smooth_hits	mass_estimate	mass_stderr
cone	−7	89	3888	3.0000	0.0481
cone	−1	83	1296	1.0000	0.0278
cone	9	73	3024	2.3333	0.0424

Graph

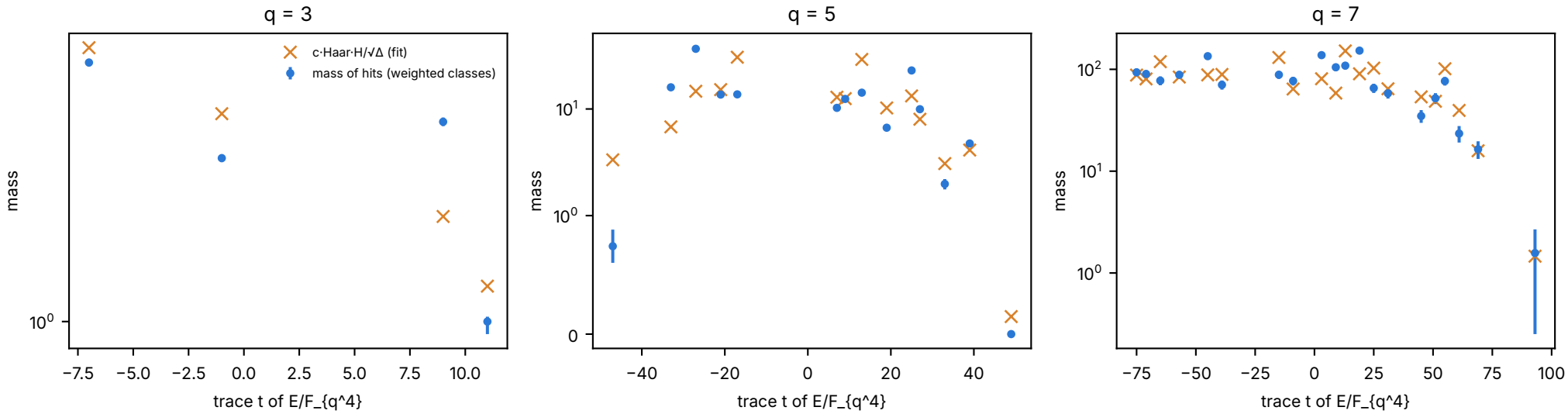


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.